

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN



BÁO CÁO ĐỀ CƯƠNG THỰC TẬP CƠ SỞ

ĐỀ TÀI:

**Nghiên cứu học đa nhiệm và cơ chế Attention trong bài toán
phân đoạn và phân loại u não từ ảnh MRI**

Giảng viên hướng dẫn: TS.Kim Ngọc Bách

Sinh viên thực hiện: Đặng Tuấn Minh

Mã Sinh Viên: B23DCCN539

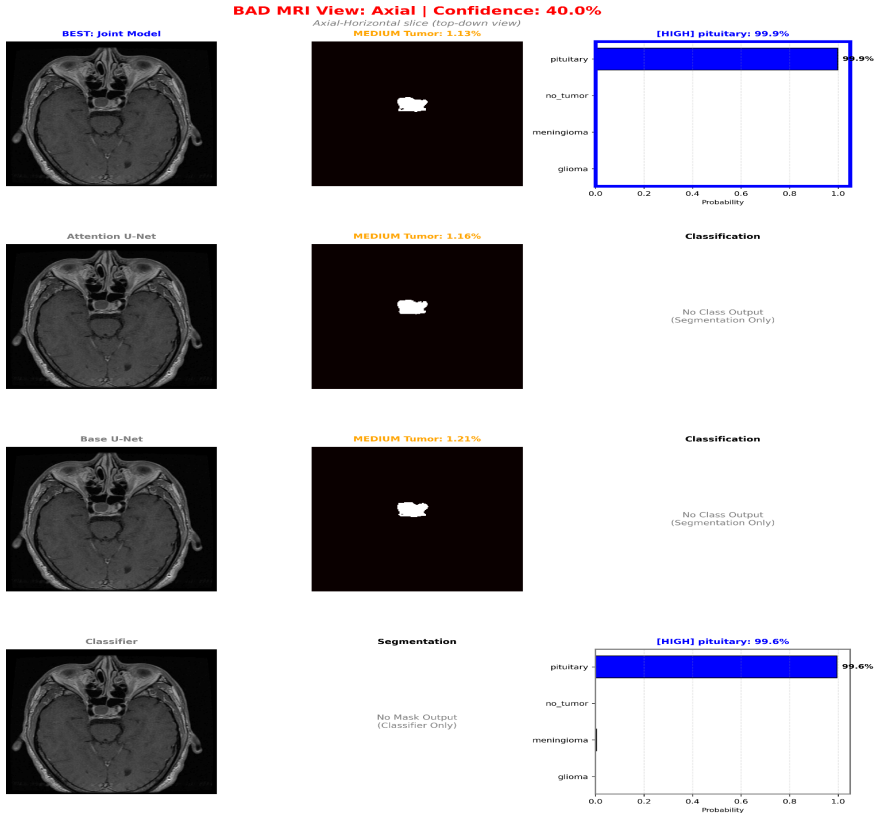
HÀ NỘI, THÁNG 5/2026

Báo cáo tuần 9

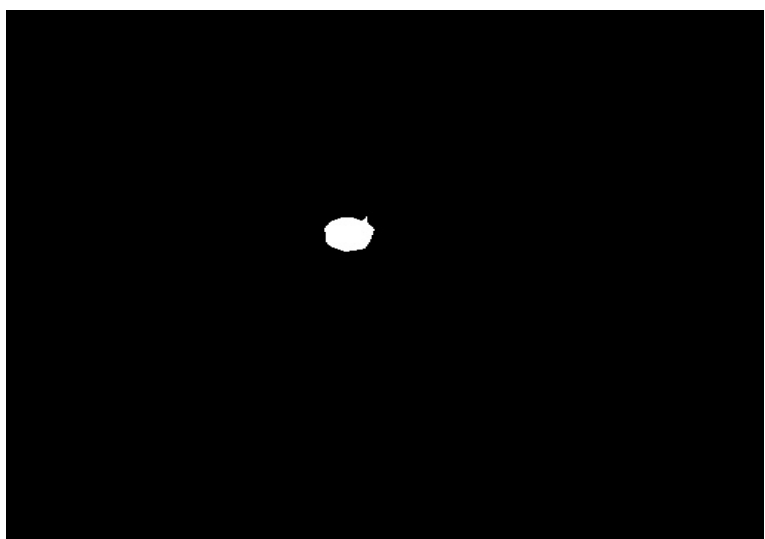
I) Công việc hoàn thành

Đã thực hiện train xong các model. Tuần này em bắt đầu tổng hợp các kiến thức để viết báo cáo cuối kỳ.

Một vài kết quả thử nghiệm:



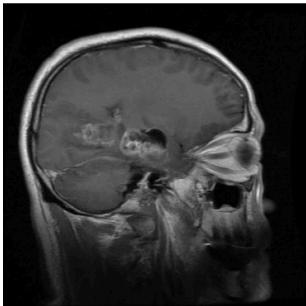
GROUND TRUTH



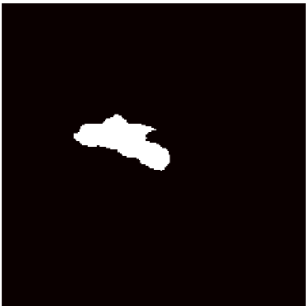
OK MRI View: Sagittal | Confidence: 66.7%

Sagittal-Side slice (left-right view)

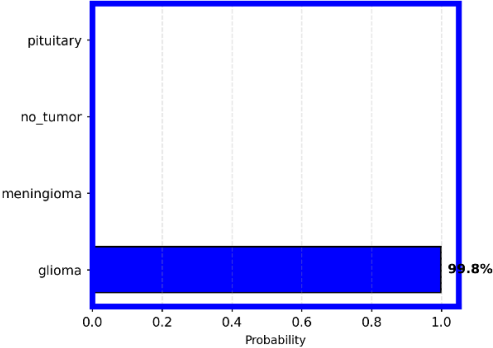
BEST: Joint Model



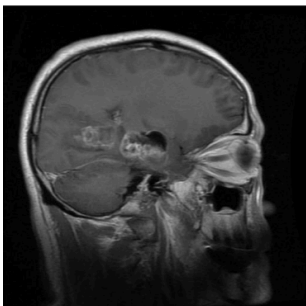
MEDIUM Tumor: 2.85%



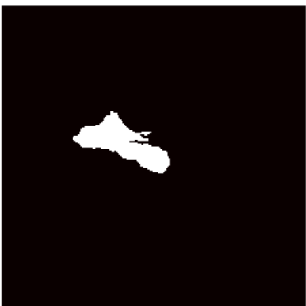
[HIGH] glioma: 99.8%



Attention U-Net



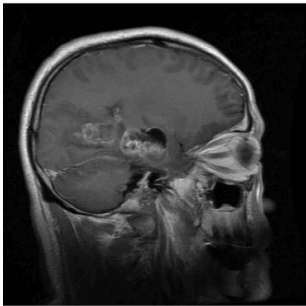
MEDIUM Tumor: 2.74%



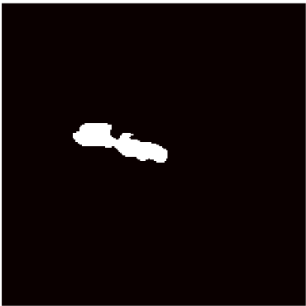
Classification

No Class Output
(Segmentation Only)

Base U-Net



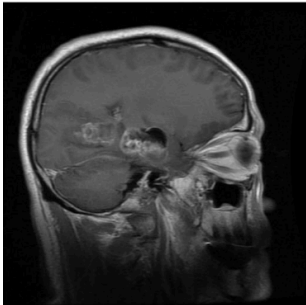
MEDIUM Tumor: 1.89%



Classification

No Class Output
(Segmentation Only)

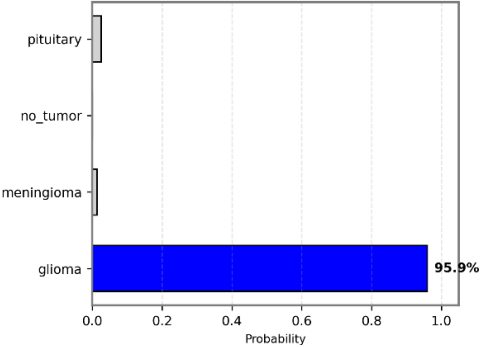
Classifier



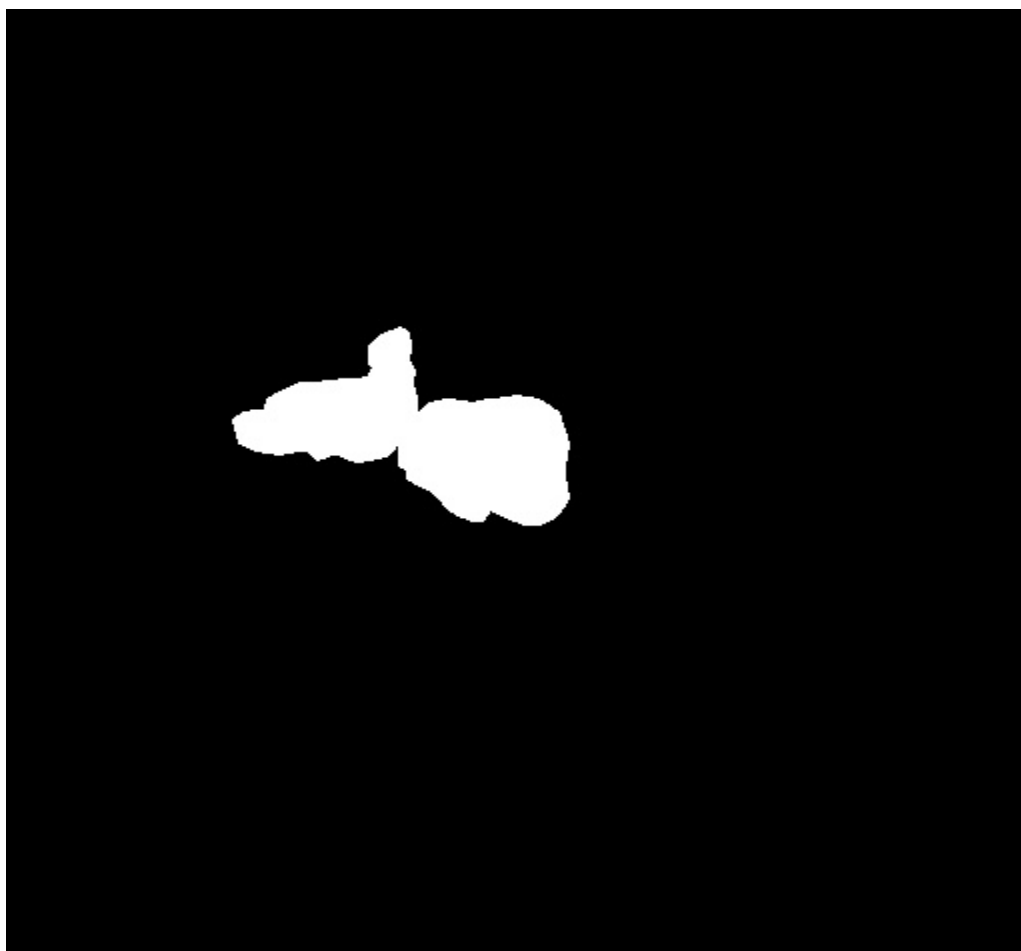
Segmentation

No Mask Output
(Classifier Only)

[HIGH] glioma: 95.9%



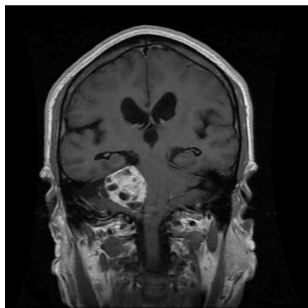
GROUND TRUTH



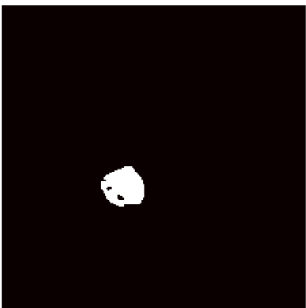
OK MRI View: Sagittal | Confidence: 66.7%

Sagittal-Side slice (left-right view)

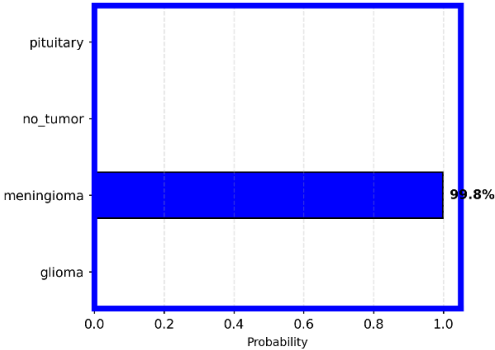
BEST: Joint Model



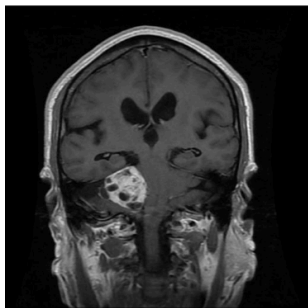
MEDIUM Tumor: 1.37%



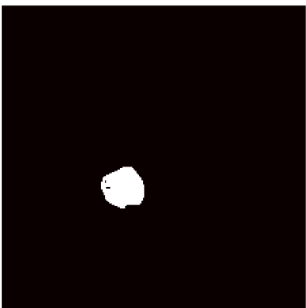
[HIGH] meningioma: 99.8%



Attention U-Net



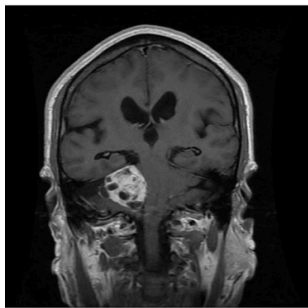
MEDIUM Tumor: 1.46%



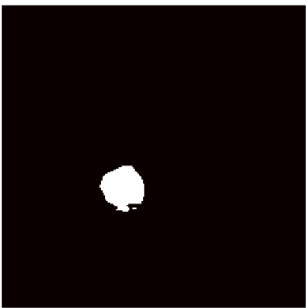
Classification

No Class Output
(Segmentation Only)

Base U-Net



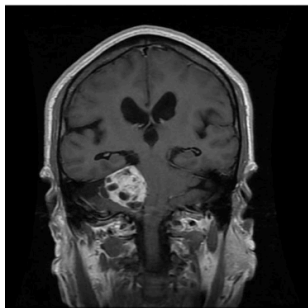
MEDIUM Tumor: 1.57%



Classification

No Class Output
(Segmentation Only)

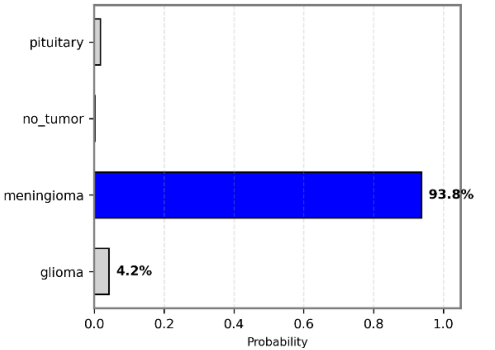
Classifier



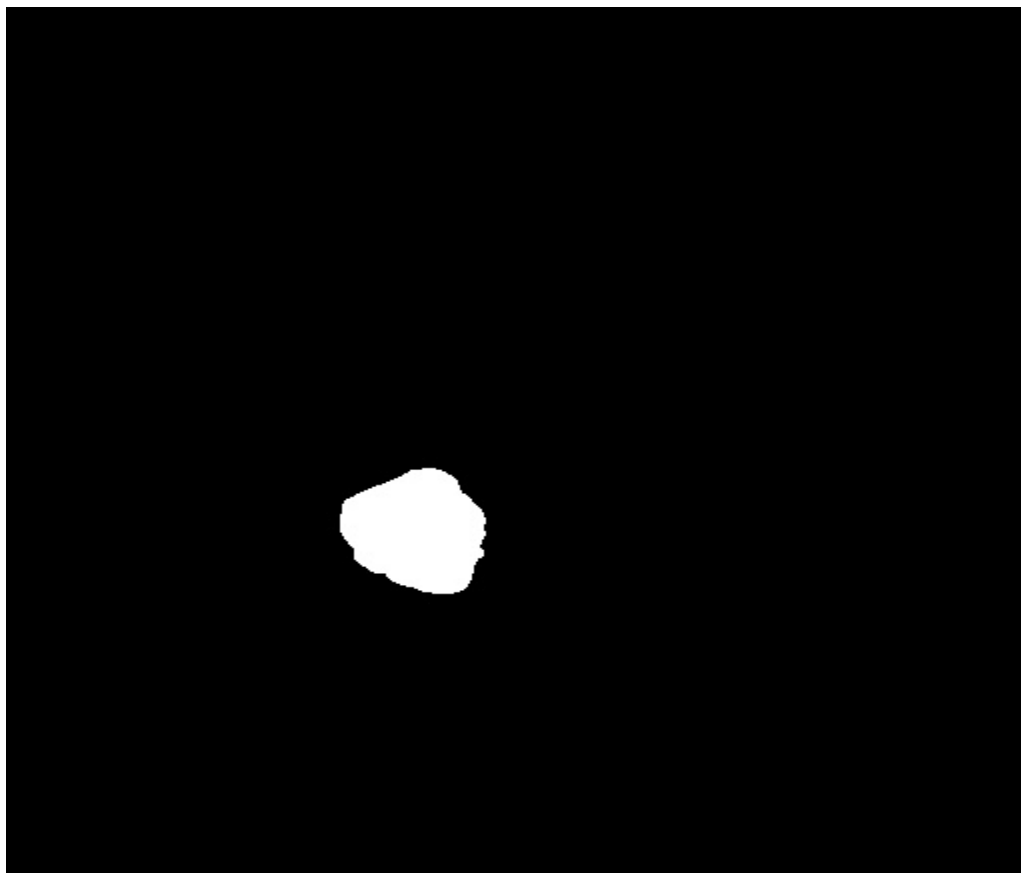
Segmentation

No Mask Output
(Classifier Only)

[HIGH] meningioma: 93.8%



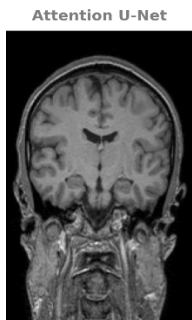
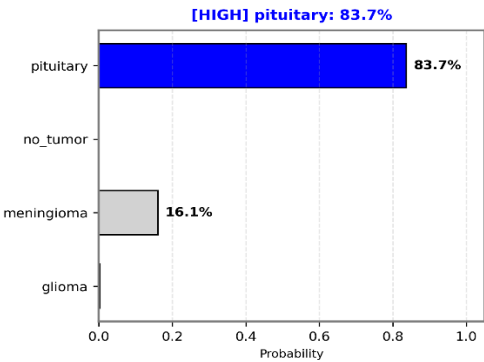
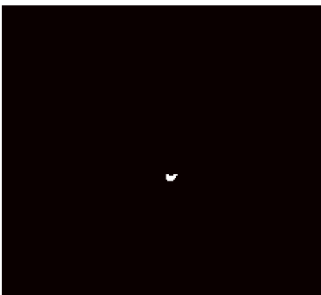
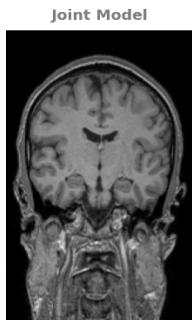
GROUND TRUTH



Trường hợp Không có khối u

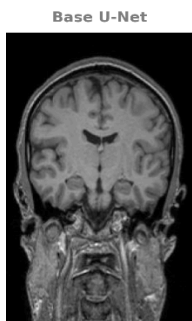
OK MRI View: Coronal | Confidence: 66.7%

Coronal-Frontal slice (front-back view)



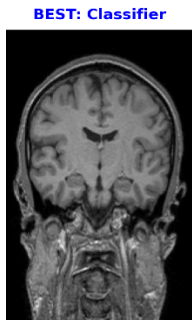
Classification

No Class Output
(Segmentation Only)



Classification

No Class Output
(Segmentation Only)



Segmentation

No Mask Output
(Classifier Only)

